

Интегрирование рациональных дробей

$$\int \frac{x dx}{(x+1)(x+2)(x+3)} \leftarrow \text{правильно}$$

лучше", где "правильности" —

$$A = \frac{x-1}{1 \cdot 2} = -\frac{1}{2}$$

$$\sim 1891 \int \frac{x dx}{(x+1)^3 (x-1)^2} = \int \frac{Ax^2 + Bx + C}{(x-1)(x+1)^2} + \int \frac{Dx + F}{(x-1)(x+1)}$$

$$\frac{x}{(x-1)^2 (x+1)^3} = \left(\frac{Ax^2 + Bx + C}{(x-1)(x+1)^2} \right)' + \frac{Dx + F}{(x-1)(x+1)}$$

$$= \begin{cases} A = \frac{1}{8} \\ B = -\frac{1}{8} \\ C = -\frac{1}{4} \\ D = 0 \\ F = -\frac{1}{8} \end{cases} = -\frac{1}{8} \frac{x^2 + x + 2}{(x-1)(x+1)^2} - \frac{1}{8} \int \frac{dx}{x^2-1} =$$

$$= -\frac{1}{16} \ln \left| \frac{x-1}{x+1} \right| + C.$$

$$\text{Другое решение: } \int \frac{x dx}{(x-1)^2 (x+1)^3} = \int \left(\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3} \right) dx$$

$$+ \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3} \Big) dx \sim$$

$$A = \left(\frac{x}{(x+1)^3} \right)'_{x=1} = \frac{(x+1)^3 - 3 \cdot x(x+1)^2}{(x+1)^6} \Big|_{x=1} = \frac{8-12}{64} = -\frac{1}{16}$$

$$D = \left(\frac{x}{(x-1)^2} \right)'_{x=-1} = \frac{(x-1)^2 - 2x(x-1)}{(x-1)^4} \Big|_{x=-1} = 4-4=0$$

$$C = \frac{1}{2!} \left(\frac{x}{(x-1)^2} \right)''_{x=-1} = \left(\frac{-x}{(x-1)^3} \right)'_{x=-1} = \frac{1}{16}$$

$$= -\frac{1}{16} \ln|x-1| + \frac{1}{16} \ln|x+1| + \frac{1}{8} \cdot \frac{1}{(x-1)} - \frac{1}{x} \cdot \left(-\frac{1}{2} \right) \cdot \frac{1}{(x+1)^2} + C$$

$$\int \frac{x^2 dx}{(x^2-3x+2)^2} = \int \frac{x^2 dx}{(x-1)^2(x-2)^2} = \int \left(\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-2} + \frac{D}{(x-2)^2} \right) dx =$$

$$B = \frac{1}{1} = 1 ; \quad D = \frac{4}{4} = 1$$

$$A = \left(\frac{x^2}{(x-2)^2} \right)' \Big|_{x=1} = 4.$$

$$C = \left(\frac{x^2}{(x-1)^2} \right)' = \frac{2x(x-1)^2 - 2(x-1)x^2}{(x-1)^4} \Big|_{x=2} = \frac{4-8}{1} = -4$$

$$= 4 \ln|x-1| - 4 \ln|x-2| - \frac{1}{x-1} - \frac{4}{x-2} + C.$$

Комплексные корни многочлена:

$$\frac{P_{k-1}(x)}{(x^2+px+q)^k} = \frac{A_1x+B_1}{x^2+px+q} + \dots + \frac{A_kx+B_k}{(x^2+px+q)^k}$$

метод residues:

$$\int \frac{u^{2k-1}}{1+u^{2k}} du = \int \sum_{n=1}^k \frac{A_n x + B_n}{u^2 - 2u \cos \alpha(2n-1) + 1}$$

$$u^{2k} = -1 = e^{i\pi(1+2\pi n)}, \quad n = \overline{0, 2k-1}$$

$$u = e^{\frac{i\pi}{2k}}$$

$$\left(u^2 - 2u \cos \alpha(2n-1) + 1 \right)$$

$$\alpha = \frac{\pi}{2k}$$

$$\int \frac{u^{2k-1}}{1+u^{2k}} du = \int \frac{u^{2k-1}}{\prod_{n=1}^k (u^2 - 2u \cos \alpha(2n-1) + 1)} du$$

$$D/3: \sim 1874.$$

$$\sim 1874. \quad \int \frac{dx}{(x+1)(x+2)^2(x+3)^3} = \int \left(\frac{A}{x+1} + \frac{B}{x+2} + \right.$$

$$\left. + \frac{C}{(x+2)^2} + \frac{D}{x+3} + \frac{E}{(x+3)^2} + \frac{F}{(x+3)^3} \right) dx.$$

$$A = \frac{1}{6}; \quad C = \frac{1}{-1} = -1. \quad F = \frac{1}{-2} = -\frac{1}{2}.$$

$$B = \left(\frac{1}{(x+1)(x+3)^3} \right)'_{x=-2} = -$$

(или Эйлера)

$$\int \frac{u^{2k-1}}{1+u^{2k}} du$$